

### 1) Install Sweet Home 3D

<https://www.sweethome3d.com/> (link for mac; unsure for windows)

### 2) Prep Your Floor Plan Image

- Sweet Home 3D's background image tool works best with PNG/JPG.
- Convert the PDF page(s) of the building plans to PNG
  - Even just taking a screen shot of the pdf works

Name them clearly (example below):

chalmers\_floor1.png

chalmers\_floor2.png

Chalmers\_floor3.png

### 3) Create a New Project

Open Sweet Home 3D

File -> New

### 4) Import the Floor Plan as a Background Image

Go to:

Plan -> Import background image

Select your floor image (Floor 1)

Sweet Home 3D will ask you to draw a line between two points, then enter the real distance.

Pick something long and clearly labeled on the plan (a long hallway or exterior wall length). Your plan includes many measured distances, so choose a clean one (e.g., a long run like ~70' or similar).

- Click point A, click point B (ends of a known wall)
- Enter the real-world length (feet/inches)

## 5) Draw the Walls (fast + accurate method)

Use the Wall tool

- Click Create walls (wall icon)
- Click to start a wall, click to end
- Double-click to finish a chain

Recommended drawing order (not strict you can do however you want)

- Exterior outline first (big perimeter)
- Main hallways second
- Big room blocks third (classrooms, offices)
- Small partitions last

Don't worry about doors/windows yet just get the structure right.

## 6) Set Wall Thickness + Height

Wall thickness: ~6 in to 10 in (depends on the plan looks like)

Adjust a specific wall height

- Select wall(s)
- Right click -> Modify walls
- Enter the custom height

## 7) Interior Design

Sweet homes have pre-made elements that it provides that can be added to the 3d model

- Using this feature you can search for anything that might help make the model look more realistic (e.g. adding desks, chairs, or sinks)

For now we have decided to just leave open holes for doors so don't use door elements that are provided.

Concerns:

- The model is mainly a blank slate so right now I'm not too sure how we are going to be able to add carpet or wall textures (might have to happen in blender)
- There are also objects and furniture in the buildings that won't have pre-existing elements to be added so we may have to do an extra scanning step (extra though)
- Photo references of the building you are working on will be very useful because the plan itself can be confusing at times

## Export as OBJ

Go to:

3D View → Export to OBJ format

## Import into Blender (recommended cleanup step)

- Even if you're going to Unity, do Blender first:
- Blender -> File -> Import -> Wavefront (.obj)
- Check scale
- Separate objects if needed (doors especially)
- Fix pivots/origins (doors need hinge pivot to animate)
- Export to FBX for Unity

## Bring into Unity

In Unity:

- Drag FBX (or OBJ) into Assets
- Put it in scene
- Add colliders:
- Walls: MeshCollider or simple BoxColliders
- Floors: MeshCollider